For teachers only. Do not give this part to students.

Research conditions, distribution and timing

This group uses the worksheet without operating the VL or receiving a lesson beforehand. Both groups have 15–20 minutes in total. Use the same T for both groups in each session, including self-study, practice and checking.

Choose a common T = _____ minutes (15–20) for both groups. Select it for the class reading pace and record actual start/end times. Checking is included; do not add a lesson beforehand.

Stage	T	T=15	T=20
A: context and supporting concept diagrams	30%	4.5 min	6 min
B1–B2: guided practice	35%	5.25 min	7 min
B3–B4: independent application	25%	3.75 min	5 min
C: check and correct	10%	1.5 min	2 min

Distribution and teacher support

1. Give out A and B at the start. Keep A available. Explain only reading, answering and timing, not digestion concepts.
2. In B1–B2, clarify wording/procedure and point to the question’ s notes or A, without giving answers or deciding for students. Log extra clarification and use the common support rules for both groups.
3. B3–B4 use the supplied notes independently, with A available. Address procedure/access needs only, without answer feedback.
4. Save the first B answers before distributing C at checking time. Correct in another colour. Give brief consistent feedback using C, keeping first and corrected answers separate.
5. Give students only A–C. D and the complete pack are for teacher printing. Verify comparable actual total time from session records, not just a screen label.

Draft worksheet feedback guide. These marks are not equivalent to VL activity marks, SPSAI or CKT. Use separate common tests to assess research outcomes.

This material follows the specified self-study comparison condition; its activity process differs from IBL.

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Full answer key and teaching limits

B1.1 • Add 2 mL bile (膽汁) to the same oil–water mixture and shake it. Write one observable prediction and give your reason.

For example: smaller, more dispersed oil droplets; the mixture may look cloudier, because bile (膽汁) causes emulsification (乳化作用). Accept other reasoned, observable predictions; do not deduct inquiry marks just for not matching a reference appearance.

B1.2 • The main comparison changes the combination of digestive juices. Classify the factors as independent, dependent or controlled variables.

Independent: combination of juices. Dependent: appearance. Controlled: starting oil/water volume, temperature, time, total added volume and shaking method.

B1.3 • Choose all the conditions needed for a fair comparison.

Choose equal time, equal starting and total added volumes, and equal shaking; do not choose extra total liquid for X + Y.

B2.1 • To compare adding digestive juice with adding only water, which of the four sets is the control group? Why keep the total volume added the same?

The 2 mL water set is the control group for this question. Equal added volume avoids a difference in total volume or dilution.

B2.2 • Use the following recipes to complete the four setup labels and added volumes.

In order: water control, 2 mL water; X, 2 mL; Y, 2 mL; X + Y, 1 mL each.

B2.3 • Someone adds 2 mL water to the control but 2 mL X and 2 mL Y to the combined set, with other conditions equal. Is this fair? Identify the problem, fix the two volumes and explain why.

Unfair: 2 mL versus 4 mL. Use 1 mL of each juice, keeping the total at 2 mL to avoid a volume/dilution difference.

B2.4 • Limits of appearance evidence: choose the best statement, then name another kind of evidence needed.

Choose the statement that appearance gives clues but does not prove glycerol (甘油) and fatty acids (脂肪酸) formed. Other chemical product tests and repeated tests are needed.

Appearance is a teaching clue: clarity alone does not prove glycerol (甘油) and fatty acids (脂肪酸) formed or measure speed. X + Y uses 1 mL each, while a single-juice set uses 2 mL. The different individual amounts cannot prove a measured combined effect or faster rate.

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Full answer key and teaching limits

B3.1 • Observe Figure 1 and record the four appearances.

Water: separated layers; X: cloudy; Y: thinner oil layer, still separated; X + Y: clear.

B3.2 • X: choose the appearance, then explain it using emulsification (乳化作用).

Cloudy emulsion (乳狀液). X disperses oil droplets by emulsification (乳化作用), a form of physical digestion (物理消化), without changing the lipid molecules.

B3.3 • Total surface area after droplets become smaller: choose the answer and explain how this helps Y act.

Increases. For the same total amount of lipid, small droplets provide more surface for lipase (脂肪酶) to act on. This is not a measurement of speed.

B4.1 • To study boiling one digestive juice, which set should be the control group? What must happen to the boiled liquid before comparison?

The unboiled X + Y set, 1 mL each; cool the boiled juice to the common 37°C and keep the other conditions equal. This control contains X and Y.

B4.2 • Observe Figure 2, record the heated sets and compare them with the unboiled X + Y control group.

Boiled X + unboiled Y: clear; unboiled X + boiled Y: cloudy.

B4.3 • Juice that loses its action after heating: choose the answer and explain why cooling does not restore its action.

Y. Lipase (脂肪酶) is a protein; boiling causes denaturation and loss of function. Cooling does not restore it in this model.

B4.4 • Use at least one main-set comparison and one heating comparison to explain emulsification (乳化作用), enzyme action and denaturation. Name the sets. Do not claim a measured combined effect or increase in speed.

For example, compare water with X for emulsification (乳化作用), and unboiled X + Y with boiled Y for enzyme loss. Explain bile (膽汁), lipase (脂肪酶) hydrolysis and heat denaturation. Appearance and different individual juice amounts limit measured claims. Other sound specific comparisons are accepted.

Appearance is a teaching clue: clarity alone does not prove glycerol (甘油) and fatty acids (脂肪酸) formed or measure speed. X + Y uses 1 mL each, while a single-juice set uses 2 mL. The different individual amounts cannot prove a measured combined effect or faster rate.

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Question-by-question mapping

Original VL item	Treatment	Revised item	Shared goal	Reason
Complete your prediction / If I add _____to the oil and water, the mixture in the test tube will _____. / My reason (required; there is no single correct answer)	Rewritten	B1.1 Add 2 mL bile (膽汁) to the same oil–water mixture and shake it. Write one observable prediction and give your reason.	Inquiry design: apply a supplied concept to write a testable prediction and reason.	Write a prediction using supplied knowledge of bile (膽汁) and emulsification (乳化作用); do not present it as a pre-learning original guess.
02 • Fair test design： Independent variable (the factor changed on purpose) / Dependent variable (the result you measure or observe) / Controlled variables (the factors kept the same)	Retained / equivalent rewrite	B1.2 The main comparison changes the combination of digestive juices. Classify the factors as independent, dependent or controlled variables.	Classification: identify the three kinds of variables.	Keep the common content and skills, practised through self-study or static data.
Which conditions does this fair test need?	Retained / equivalent rewrite	B1.3 Choose all the conditions needed for a fair comparison.	Inquiry design and experimental reasoning: identify sound fixed conditions.	Keep the common content and skills, practised through self-study or static data.
Does this experiment need a control group? If so, how would you design the setups to compare?	Rewritten	B2.1 To compare adding digestive juice with adding only water, which of the four sets is the control group? Why keep the total volume added the same?	Inquiry design: explain the purpose of a control.	Keep the common content and skills, practised through self-study or static data.
Written design (can replace a drawing): describe the four setups, how much liquid you add and which conditions you keep the same. / Draw and label your setup / Use your experimental variables to help you draw and label your setup.	Rewritten	B2.2 Use the following recipes to complete the four setup labels and added volumes.	Experimental reasoning and communication: read and label setups and volumes.	Label and evaluate a given procedure while practising apparatus and fair-comparison skills.

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Question-by-question mapping

Original VL item	Treatment	Revised item	Shared goal	Reason
Written design (can replace a drawing): describe the four setups, how much liquid you add and which conditions you keep the same. / Does this experiment need a control group? If so, how would you design the setups to compare?	Rewritten	B2.3 Someone adds 2 mL water to the control but 2 mL X and 2 mL Y to the combined set, with other conditions equal. Is this fair? Identify the problem, fix the two volumes and explain why.	Inquiry design and experimental reasoning: evaluate and correct an unfair procedure.	Keep the common content and skills, practised through self-study or static data.
3. What can this experiment tell us?：We only observed the oil layer and whether the mixture was cloudy or clear. We did not test for digestion products. Which statement fits best?_____	Retained / equivalent rewrite	B2.4 Limits of appearance evidence: choose the best statement, then name another kind of evidence needed.	Inference and communication: distinguish clues, direct evidence and limits.	Keep the common content and skills, practised through self-study or static data.
Your first observation：Describe only what you can see in the oil + water test tube (do not yet guess what X and Y are or what the results will be). / Select the appearance based on what you observe:	Merged / static data	B3.1 Observe Figure 1 and record the four appearances.	Observation and communication: distinguish the four appearances.	Remove the separate initial-observation item; retain observation evidence in the four main and two heating records.
1. What does digestive juice X do?：Digestive juice X mainly causes lipids to form _____.	Retained / equivalent rewrite	B3.2 X: choose the appearance, then explain it using emulsification (乳化作用).	Inference: link an appearance with emulsification (乳化作用).	Keep the common content and skills, practised through self-study or static data.
2. How does digestive juice X help digestive juice Y act?：Digestive juice X first breaks up large oil droplets into small droplets, causing their total surface area to _____, which helps digestive juice Y act.	Retained / equivalent rewrite	B3.3 Total surface area after droplets become smaller: choose the answer and explain how this helps Y act.	Inference: explain surface area without inventing speed data.	Keep the common content and skills, practised through self-study or static data.

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Question-by-question mapping

Original VL item	Treatment	Revised item	Shared goal	Reason
Does this experiment need a control group? If so, how would you design the setups to compare? / Which digestive juice is affected by high temperature?	Rewritten	B4.1 To study boiling one digestive juice, which set should be the control group? What must happen to the boiled liquid before comparison?	Inquiry design and experimental reasoning: choose the control for the question and control reaction temperature.	Keep the common content and skills, practised through self-study or static data.
Extension: the effects of heating on digestive juices：Which digestive juice is affected by high temperature?	Static data	B4.2 Observe Figure 2, record the heated sets and compare them with the unboiled X + Y control group.	Observation: compare heating treatments with a control group.	Keep the common content and skills, practised through self-study or static data.
Extension conclusion: the digestive juice that loses its function after boiling is_____.	Retained / equivalent rewrite	B4.3 Juice that loses its action after heating: choose the answer and explain why cooling does not restore its action.	Inference and content knowledge: explain heat treatment using enzyme denaturation.	Keep the common content and skills, practised through self-study or static data.
Do your observations support your first idea to test? Name the groups you compared and what you saw. Use the key learning points to improve your explanation.	Rewritten	B4.4 Use at least one main-set comparison and one heating comparison to explain emulsification (乳化作用), enzyme action and denaturation. Name the sets. Do not claim a measured combined effect or increase in speed.	Communication and content knowledge: explain concepts using named comparisons.	Concepts are taught first; assess their use with evidence rather than inventing an original hypothesis.

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Worksheet feedback guide

Draft worksheet feedback guide. These marks are not equivalent to VL activity marks, SPSAI or CKT. Use separate common tests to assess research outcomes.

Item	Maximum	Shared skill / knowledge	Marking criteria
B1.1	2	Inquiry design: apply a supplied concept to write a testable prediction and reason.	1 for an observable prediction with a clear treatment; 1 for a sound reason linked to emulsification (乳化作用). Do not score by guessing the result.
B1.2	4	Classification: identify the three kinds of variables.	Independent 1, dependent 1; five controlled factors at 0.4 each, total 2. A factor from a wrong category gives 0 for that category.
B1.3	1	Inquiry design and experimental reasoning: identify sound fixed conditions.	1 for all three correct conditions with no wrong choice; 0 for another attempted set.
B2.1	2	Inquiry design: explain the purpose of a control.	1 for the control group, 1 for a sound reason to keep total volume fixed.
B2.2	2	Experimental reasoning and communication: read and label setups and volumes.	2 for all four correct setup/volume labels; 0.5 per complete label.
B2.3	3	Inquiry design and experimental reasoning: evaluate and correct an unfair procedure.	1 for the flaw, 1 for corrected volumes, 1 for the fairness explanation.
B2.4	2	Inference and communication: distinguish clues, direct evidence and limits.	1 for the sound statement, 1 for a useful additional test or repeated evidence.
B3.1	4	Observation and communication: distinguish the four appearances.	1 per correctly described set, total 4.

B first-answer practice maximum: 34. Use C corrections for separate feedback; do not overwrite the first-answer marks.

Mark missing answers as unavailable and follow the study’ s missing-data rule. Do not invent evidence. Judge explanations and links to data, not keywords.

These marks are not equivalent to VL original-prediction, virtual-operation or completion marks and must not be converted to its 32-point score. Assess research outcomes using separate common tests.

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Worksheet feedback guide

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Item	Maximum	Shared skill / knowledge	Marking criteria
B3.2	2	Inference: link an appearance with emulsification (乳化作用).	1 for the appearance; 1 for a sound explanation of emulsification (乳化作用) without molecular change.
B3.3	2	Inference: explain surface area without inventing speed data.	1 for increase, 1 for explaining the available surface and enzyme action.
B4.1	2	Inquiry design and experimental reasoning: choose the control for the question and control reaction temperature.	1 for the unboiled control group, 1 for cooling to the common temperature for a fair comparison.
B4.2	2	Observation: compare heating treatments with a control group.	1 for each set.
B4.3	2	Inference and content knowledge: explain heat treatment using enzyme denaturation.	1 for Y, 1 for protein-enzyme denaturation and why cooling alone does not restore it.
B4.4	4	Communication and content knowledge: explain concepts using named comparisons.	1 each for emulsification (乳化作用), enzyme hydrolysis and denaturation; 1 for clear main/heating comparisons. Judge explanations, not keywords.

B first-answer practice maximum: 34. Use C corrections for separate feedback; do not overwrite the first-answer marks.

Mark missing answers as unavailable and follow the study’s missing-data rule. Do not invent evidence. Judge explanations and links to data, not keywords.

These marks are not equivalent to VL original-prediction, virtual-operation or completion marks and must not be converted to its 32-point score. Assess research outcomes using separate common tests.